P3.py

```
import pandas  as pd


data = pd.DataFrame({
	"info": [
		"name=kamal mobile=8369084928 college=rait",
		"name=aryaman college=dmcoe mobile=8369084968",
		"mobile=7369084928 college=wattumul name=tushar",
		"name=pranav mobile=83677084928 college=dmcoe",
	]
})

print(data)
data["name"] = data["info"].str.extract(r'name=(\w+)')
data["college"] = data["info"].str.extract(r'college=(\w+)')
data["phone"] = data["info"].str.extract(r'mobile=(\d+)')

print(data)
```

P4.py

```
# import lib
import pandas as pd
from sklearn.linear_model import LinearRegression

# load data
data = pd.read_csv("area_location_price.csv")
print(data)

# clean data
def clean_info(data):
	data["area"] = data["info"].str.extract(r'area=(\d+)')
	data["location"] = data["info"].str.extract(r'location=(\w+)')
	data = pd.get_dummies(data, columns=["location"], dtype=int)
	if "location_karjat" not in data.columns:
		data["location_karjat"] = 0
	if "location_khandala" not in data.columns:
		data["location_khandala"] = 0
	if "location_lonavala" not in data.columns:
		data["location_lonavala"] = 0
	return data[["area", "location_karjat",  "location_khandala", "location_lonavala"]]

# features and target
features = clean_info(data)
print(features)
target = data["price"]

# model
model = LinearRegression()
model.fit(features, target)

# prediction
area = int(input("enter area in square feet "))
location = input("enter location: karjat / khandala / lonavala ").lower()
test = pd.DataFrame({"info": ["area=" + str(area) + " location=" + str(location)]})
price = model.predict(clean_info(test))
print(price)
```